

Assignment 2

Instructions:

- Show all steps — all derivations, intermediate calculations, and reasoning must be clearly written.
- Round all numeric answers to 3 decimal places unless otherwise specified.
- Use proper notation for summations, derivatives, and vectors throughout your work.
- State any assumptions you make.
- For Problem 6, you may copy the diagrams on paper and draw the decision boundary on them.
- Plagiarism is strictly prohibited. Any copied work from other students or sources will receive no marks. There is zero tolerance for academic dishonesty. All submissions must be your own work.

Problem 1

Consider the following 1-dimensional training set:

x	y
-2	7
-1	4
0	3
1	4
2	7

Our goal is to find a regression model that maps x to the target value y . We use a linear regression model and assume the relationship

$$y = w_1x + w_0 + \varepsilon$$

$$= w^T \phi(x) + \varepsilon$$

where the feature mapping is defined as

$$\phi(x) = \begin{bmatrix} 1 \\ x \end{bmatrix}.$$

This feature mapping allows us to absorb the bias term w_0 into the vector w . Using this feature mapping, the design matrix can be written as

$$X = [\phi(x_1) \ \phi(x_2) \ \dots \ \phi(x_n)].$$

Using the formula from class, compute the closed-form solution for w .

Problem 2

Suppose that we are given

$$f(x) = x^3 + x^2$$

and learning rate

$$\alpha = \frac{1}{4}.$$

(a)

First, write down the update rule for gradient descent in general and then for this specific function.

(b)

Write down one condition to terminate the gradient descent algorithm (in general).

Problem 3

The following measurements have been obtained in a study:

y_i	x_1	x_2
10	5	8
30	20	2
20	5	9
25	15	11
15	10	10

Assuming a multivariate linear regression model:

$$y_i = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

(a) Matrix Formulation

Construct the design matrix X and response vector Y for the given data. Then compute the matrices $X^T X$ and $X^T Y$ required for the closed-form solution of the multivariate linear regression model. Finally, compute $(X^T X)^{-1}$.

(b) Estimation of Parameters

Using the closed-form solution

$$\hat{\theta} = (X^T X)^{-1} X^T Y,$$

compute the parameter vector. $\hat{\theta}$. Write the estimated regression equation and interpret the coefficients.

(c) Prediction

Using the estimated regression model from part (b), predict the value of y for a new observation with $x_1 = 5$ and $x_2 = 8$.

Problem 4

Consider the following dataset with 2 features (x_1, x_2) and a target y :

x_1	x_2	y
1	2	4
2	0	3
3	1	5

We want to fit a linear regression model:

$$y = w_0 + w_1 x_1 + w_2 x_2$$

using gradient descent. The cost function is:

$$J(w_0, w_1, w_2) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

(a) Gradient Descent Update

Use gradient descent with learning rate $\alpha = 0.1$ and initial weights $w_0 = w_1 = w_2 = 0$.

1. Write the gradient descent update rules for w_0, w_1, w_2 .
2. Compute the gradient and update the weights after one iteration using all data points (batch gradient descent).

(b) Compute Loss

1. Compute the loss with the initial weights.
2. Compute the loss after the first weight update.

(c) Normal Equation

Compute the weights using the normal equation:

$$\mathbf{w} = (X^T X)^{-1} X^T y$$

where X is the design matrix with a column of ones for the bias term.

(d) Comparison

Compare the weights obtained via gradient descent (after one iteration) with the weights from the normal equation. Discuss the effect of the first gradient descent update on the loss.

Problem 5

Explain whether each scenario is a classification or regression problem.

(a)

We collect a set of data on the 100 highest paid professional basketball players. For each player we record salary, height, weight, age, number of games played in their career, points scored, assists, rebounds, and blocks. We are interested in understanding which factors affect player salary.

(b)

We would like to predict whether FAST will win its next inter-university coding competition against NUST. To do this, we collect data from the last 30 competitions in which FAST and NUST competed against each other. For each competition, we record the outcome (whether FAST won or lost), whether the event was hosted at FAST, NUST, or a neutral venue, the average CGPA of the top five team members from each university, the number of ICPC-experienced participants on each team, the average Codeforces rating of the team members, the number of problems solved in their previous competition, the average time taken to solve problems during the contest, the overall difficulty level of the competition, the number of participants with prior national-level experience, and 12 additional performance-related variables. Using this information, we aim to build a model that predicts whether FAST will win the next coding competition against NUST.

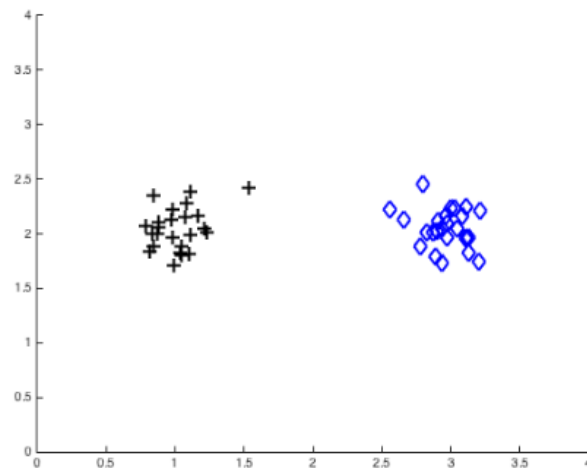
(c)

We are interested in predicting rainfall in Palo Alto based on rainfall in other parts of the world. For each week of the past year, we record the number of inches of rainfall in Palo Alto, New York City, Longyearbyen, London, Beijing, Sydney, Cairo, and Buenos Aires.

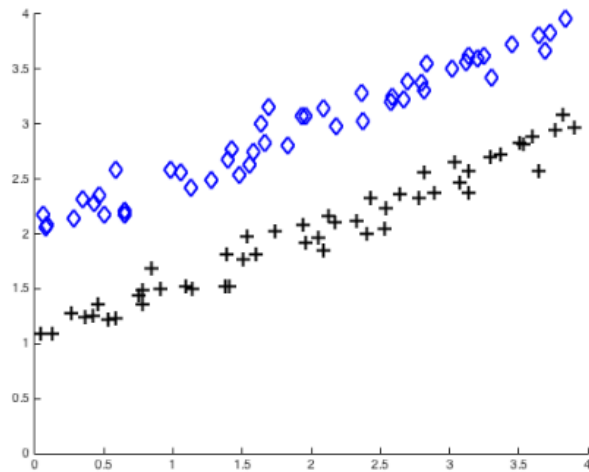
Problem 6

Consider the problem of classifying the following 2-dimensional datasets into two classes (represented by 'plus' and 'diamond') using Logistic Regression. For each of the three following datasets, draw the Logistic Regression decision boundary with a line. If the classifier cannot classify the data correctly, write one sentence explaining why not. Note: You need to draw decision boundary only for the cases where the data can be classified correctly.

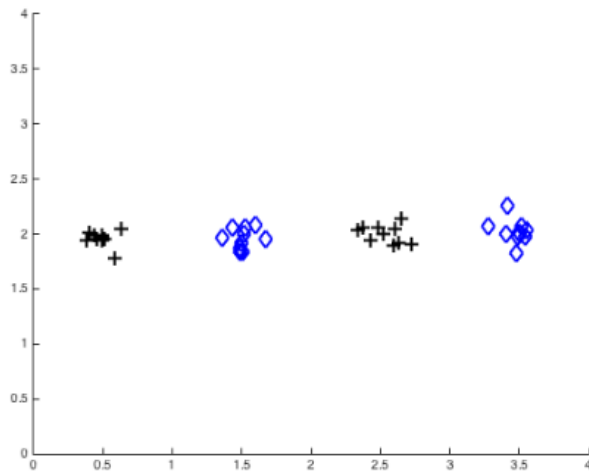
(a)



(b)



(c)



Problem 7

Alia is in her 8th semester at NUCES and wants to register for a popular course using Flex, the online student portal. In the past, Flex has sometimes been slow or unresponsive. She wants to estimate the probability that she will successfully log in and register. Historical data from the past 7 semesters has been collected:

Semester	TimeTaken (s)	CoursePopularity	Success
1	5	30	1
2	12	80	0
3	7	25	1
4	10	90	0
5	6	70	1
6	14	100	0
7	8	40	1

We fit a logistic regression model with the following weights (already trained):

$$\theta_0 = 2.5, \quad \theta_1 = -0.3, \quad \theta_2 = -0.02$$

The logistic regression model is:

$$\log \frac{p}{1-p} = \theta_0 + \theta_1 \cdot \text{TimeTaken} + \theta_2 \cdot \text{CoursePopularity}$$

where $p = P(\text{Success} = 1)$.

Questions

1. Using the given weights, compute the probability that Alia successfully registers if:

$$\text{TimeTaken} = 8, \quad \text{CoursePopularity} = 85$$

2. Compute the probability if TimeTaken improves to 6 seconds but CoursePopularity stays the same.
3. Interpret the effect of each coefficient (θ_1 and θ_2) on Alia's chances of successfully registering.
4. Discuss the reliability of this model given only 7 past semesters of data and suggest ways Alia could improve her prediction for future registration days.

Problem 8

Use the same dataset from the previous problem:

Semester	TimeTaken (s)	CoursePopularity	Success
1	5	30	1
2	12	80	0
3	7	25	1
4	10	90	0
5	6	70	1
6	14	100	0
7	8	40	1

Consider the logistic regression model:

$$\hat{y}_i = \sigma(\theta_0 + \theta_1 \cdot \text{TimeTaken}_i + \theta_2 \cdot \text{CoursePopularity}_i), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

with initial weights:

$$\theta_0 = 2.5, \quad \theta_1 = -0.3, \quad \theta_2 = -0.02$$

and learning rate $\alpha = 0.1$.

Questions

1. Compute the predicted probabilities $\hat{y}_i$ for all 7 semesters using the current weights.
2. Compute the gradients for $\theta_0, \theta_1, \theta_2$ using:

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij}, \quad j = 0, 1, 2$$

where $x_{i0} = 1$ for the intercept, $x_{i1} = \text{TimeTaken}_i$, $x_{i2} = \text{CoursePopularity}_i$.

3. Perform one gradient descent update for each weight:

$$\theta_j := \theta_j - \alpha \frac{\partial J}{\partial \theta_j}, \quad j = 0, 1, 2$$